

Lie Groups and Lie Algebras

Ashan Jayamal

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Chapter 1

Notation and Some definitions

1. $GL_n(\mathbb{C})$: the group of all $n \times n$ invertible matrices with entries in the field $\mathbb{F}$.
2. $M_n(\mathbb{C})$: the set of all $n \times n$ matrices with entries in the field $\mathbb{F}$.
3. $U(n)$: $n \times n$ unitary group
4. SU : Special Unitary Group (Unitary Group + $\det=1$)
5. $O(n)$: $n \times n$ Orthogonal group
6. $SO(n)$: $n \times n$ Special orthogonal Group (Orthogonal Group + $\det=1$)

Chapter 2

Lie Group

Definition 2.1 (Lie Group). A *Lie group* is a smooth manifold G which is also a group, such that the group multiplication

$$\mu : G \times G \rightarrow G, \quad (g, h) \mapsto gh$$

and the inverse map

$$\iota : G \rightarrow G, \quad g \mapsto g^{-1}$$

are both smooth.

Example 2.1. Let $G = \mathbb{R} \times \mathbb{R} \times S$, equipped with the group operation

$$(x_1, y_1, \lambda_1) \cdot (x_2, y_2, \lambda_2) = (x_1 + x_2, y_1 + y_2, e^{ix_1 y_2} \lambda_1 \lambda_2).$$

Then G is a Lie group.

Definition 2.2. Let A_m be a sequence of complex matrices in $\mathbb{M}_n(\mathbb{C})$. We say that $A_m \rightarrow A$ (i.e., A_m converges to a matrix A) if each entry of A_m converges to the corresponding entry of A as $m \rightarrow \infty$; In other words,

$$(A_m)_{jk} \rightarrow A_{jk} \quad \text{for all } 1 \leq j, k \leq n.$$

Definition 2.3 (Matrix Lie Group). A *matrix Lie group* is a subgroup $G \subseteq \text{GL}(n, \mathbb{C})$ with the following property: If $\{A_m\}$ is any sequence of matrices in G such that $A_m \rightarrow A$, then either $A \in G$ or A is not invertible.

Example 2.2 (A Subgroup of $\text{GL}(2, \mathbb{C})$ That Is Not a Matrix Lie Group). Let $\alpha \in \mathbb{R} \setminus \mathbb{Q}$, and define the subgroup $G \subseteq \text{GL}(2, \mathbb{C})$ consisting of matrices of the form

$$G = \left\{ \begin{pmatrix} e^{it} & 0 \\ 0 & e^{i\alpha t} \end{pmatrix} : t \in \mathbb{R} \right\}.$$

Clearly, G is a subgroup of $\mathrm{GL}(2, \mathbb{C})$. The closure of G in the standard topology is the torus subgroup

$$\overline{G} = \left\{ \begin{pmatrix} e^{i\theta} & 0 \\ 0 & e^{i\phi} \end{pmatrix} : \theta, \phi \in \mathbb{R} \right\} = \left\{ \begin{pmatrix} z_1 & 0 \\ 0 & z_2 \end{pmatrix} : |z_1| = |z_2| = 1 \right\},$$

which is a compact Lie group. Since G is not closed, it fails the closure condition required for matrix Lie groups. Thus, G is not a matrix Lie group.

This subgroup G is sometimes referred to as an *irrational line in a torus*.

Example 2.3 (Example of Matrix Linear Groups).

1.

Definition 2.4 (Lie Group homomorphism and Isomorphisms). Let G and H be matrix Lie groups. A map $\varphi : G \rightarrow H$ is called a *Lie group homomorphism* if

- i. φ is a group homomorphism, and
- ii. φ is continuous.

If, in addition, φ is one-to-one and onto, and the inverse map φ^{-1} is continuous, then φ is called a *Lie group isomorphism*.

Example 2.4 (Examples of Lie Group Homomorphisms).

- 1. The determinant is a homomorphism $\det : \mathrm{GL}_n(\mathbb{C}) \rightarrow \mathbb{C} \setminus \{0\}$
- 2. The map $\varphi : \mathbb{R} \rightarrow \mathrm{SO}(2)$, defined by

$$\varphi(t) = \begin{pmatrix} \cos t & -\sin t \\ \sin t & \cos t \end{pmatrix}$$

is also a group homomorphism.

Chapter 3

Lie Algebra

Definition 3.1 (Lie Algebra). Let $\mathbb{F}$ be a field. A *Lie algebra* over $\mathbb{F}$ is a vector space $\mathfrak{g}$ over $\mathbb{F}$ together with a map $[\cdot, \cdot] : \mathfrak{g} \times \mathfrak{g} \rightarrow \mathfrak{g}$, called the *Lie bracket*, satisfying the following properties:

1. **Bi-linearity:** The bracket $[\cdot, \cdot]$ is bilinear.
2. **Skew-symmetry:** For all $X, Y \in \mathfrak{g}$,

$$[X, Y] = -[Y, X].$$

3. **Jacobi identity:** For all $X, Y, Z \in \mathfrak{g}$,

$$[X, [Y, Z]] + [Y, [Z, X]] + [Z, [X, Y]] = 0.$$

Two elements $X, Y \in \mathfrak{g}$ are said to *commute* if $[X, Y] = 0$. The Lie algebra $\mathfrak{g}$ is called *commutative* (or *abelian*) if $[X, Y] = 0$ for all $X, Y \in \mathfrak{g}$.

Example 3.1. Let $\mathfrak{g} = \mathbb{R}^3$ and define the bracket

$$[x, y] = x \times y, \quad x, y \in \mathbb{R}^3,$$

where $x \times y$ denotes the cross product. Then $\mathfrak{g}$ is a Lie algebra.

Proof. Bilinearity and skew-symmetry are standard properties of the cross product. To verify the Jacobi identity, it suffices (by bilinearity) to check it when

$$x = e_j, \quad y = e_k, \quad z = e_\ell,$$

where e_1, e_2, e_3 are the standard basis vectors of $\mathbb{R}^3$.

- If j, k, ℓ are all equal, each term in the Jacobi identity is zero.

- If j, k, ℓ are all distinct, the cross product of any two of e_j, e_k, e_ℓ is a multiple of the third, so again each term in the Jacobi identity is zero.
- It remains to consider the case in which two of j, k, ℓ are equal and the third is different. By reordering the terms in the Jacobi identity as necessary, it suffices to verify

$$[e_j, [e_j, e_k]] + [e_j, [e_k, e_j]] + [e_k, [e_j, e_j]] = 0. \quad (3.1)$$

The first two terms in (3.1) are negatives of each other, and the third is zero.

Thus the Jacobi identity holds, and $\mathfrak{g}$ is a Lie algebra. $\square$

Example 3.2. Let A be an associative algebra and let $\mathfrak{g}$ be a subspace of A such that

$$XY - YX \in \mathfrak{g} \quad \text{for all } X, Y \in \mathfrak{g}.$$

Then $\mathfrak{g}$ is a Lie algebra with bracket operation defined by

$$[X, Y] = XY - YX.$$

Proof. It is very easy to show the bilinear and skew symmetry properties. So, just go for Jacobi Identity.

Let $X, Y, Z \in \mathfrak{g}$, consider

$$\begin{aligned} [X, [Y, Z]] &= X(YZ - ZY) - (YZ - ZY)X = XYZ - XZY - YZX + ZYX, \\ [Y, [Z, X]] &= Y(ZX - XZ) - (ZX - XZ)Y = YZX - YXZ - ZXY + XZY, \\ [Z, [X, Y]] &= Z(XY - YX) - (XY - YX)Z = ZXY - ZYX - XYZ + YXZ. \end{aligned}$$

Adding these three expressions together, every monomial in X, Y, Z appears exactly twice, once with a plus sign and once with a minus sign:

$$\begin{aligned} (XYZ - XZY) + (XZY - XZY) + (YZX - YZX) + (YXZ - YXZ) \\ + (ZXY - ZXY) + (ZYX - ZYX) = 0. \end{aligned}$$

$\square$

Chapter 4

Matrix Exponential

The exponential of a matrix plays a crucial role in the theory of Lie groups. The exponential enters into the definition of the Lie algebra of a matrix Lie group and is the mechanism for passing information from the Lie algebra to the Lie group.

Definition 4.1 (Matrix Exponential). If X is an $n \times n$ matrix, we define the exponential of X , denoted e^X or $\exp(X)$, by the usual power series

$$e^X = \sum_{m=0}^{\infty} \frac{X^m}{m!}, \quad (4.1)$$

where $X^0 = I$, and X^m denotes the repeated matrix product of X with itself.

Proposition 4.1. *The series (4.1) converges for all $X \in M_n(\mathbb{C})$, and e^X is a continuous function of X .*

Theorem 4.1. *Let $X, Y \in M_n(\mathbb{C})$. Then the following properties hold:*

1. $e^0 = I$.
2. $e^{X^T} = (e^X)^T$, $(e^X)^* = e^{X^*}$.
3. e^X is invertible and $(e^X)^{-1} = e^{-X}$.
4. If A is invertible $n \times n$ matrix, then $e^{AXA^{-1}} = Ae^XA^{-1}$.
5. $e^{(\alpha+\beta)X} = e^{\alpha X}e^{\beta X}$ for all $\alpha, \beta \in \mathbb{C}$.
6. If $XY = YX$, then $e^{X+Y} = e^Xe^Y = e^Ye^X$.
7. If $C \in GL_n(\mathbb{C})$, then $e^{CXC^{-1}} = Ce^XC^{-1}$.

Proof. Point (1) is immediate from the definition of the exponential series. Point (2) follows by taking term-by-term adjoints in the series for e^X . Points (3) and (4) are special cases of Point (5).

To verify Point (5), note that both series e^X and e^Y converge absolutely, so we may multiply them term by term:

$$e^X e^Y = \sum_{m=0}^{\infty} \sum_{k=0}^m \frac{X^k}{k!} \frac{Y^{m-k}}{(m-k)!} = \sum_{m=0}^{\infty} \frac{1}{m!} \sum_{k=0}^m \binom{m}{k} X^k Y^{m-k}. \quad (4.2)$$

If X and Y commute, then

$$(X + Y)^m = \sum_{k=0}^m \binom{m}{k} X^k Y^{m-k},$$

so (4.2) becomes

$$e^X e^Y = \sum_{m=0}^{\infty} \frac{(X + Y)^m}{m!} = e^{X+Y}.$$

For Point (6), observe that

$$(CXC^{-1})^m = CX^mC^{-1},$$

so term by term we have

$$e^{CXC^{-1}} = \sum_{m=0}^{\infty} \frac{(CXC^{-1})^m}{m!} = \sum_{m=0}^{\infty} \frac{CX^mC^{-1}}{m!} = C \left(\sum_{m=0}^{\infty} \frac{X^m}{m!} \right) C^{-1} = Ce^XC^{-1}.$$

□

4.1 Computing Exponential

If X is diagonalizable, then there exist invertible A and $\lambda_1, \dots, \lambda_n$ such that

$$X = A \begin{bmatrix} \lambda_1 & 0 & \dots & 0 \\ 0 & \lambda_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \lambda_n \end{bmatrix} A^{-1}$$

Then by Theorem 4.1 fourth property, we get the following:

$$e^X = A \begin{bmatrix} e^{\lambda_1} & 0 & \dots & 0 \\ 0 & e^{\lambda_2} & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & e^{\lambda_n} \end{bmatrix} A^{-1}$$

If X is nilpotent then the series that defines e^X terminates.

Example 4.1. Consider the matrices

$$X_1 = \begin{bmatrix} 0 & -a \\ a & 0 \end{bmatrix}, \quad X_2 = \begin{bmatrix} 0 & a & b \\ 0 & 0 & c \\ 0 & 0 & 0 \end{bmatrix}, \quad X_3 = \begin{bmatrix} a & b \\ 0 & a \end{bmatrix}.$$

Then

$$e^{X_1} = \begin{bmatrix} \cos a & -\sin a \\ \sin a & \cos a \end{bmatrix}, \text{ and } e^{X_2} = \begin{bmatrix} 1 & a & b + \frac{ac}{2} \\ 0 & 1 & c \\ 0 & 0 & 1 \end{bmatrix}, \text{ and } e^{X_3} = \begin{bmatrix} e^a & e^a b \\ 0 & e^a \end{bmatrix}.$$

Proof.

$$e^{X_1} = \begin{bmatrix} 1 & i \\ i & 1 \end{bmatrix} \begin{bmatrix} e^{-ia} & 0 \\ 0 & e^{ia} \end{bmatrix} \frac{1}{2} \begin{bmatrix} 1 & -i \\ -i & 1 \end{bmatrix}.$$

□

Chapter 5

The Lie Algebra of a Matrix Lie Group

Now we going to discuss the Lie Algebra $\mathfrak{g}$ that associated with Matrix Lie group G .

Definition 5.1. If $G \subset GL(n, \mathbb{C})$ is a matrix Lie group, then the Lie algebra $\mathfrak{g}$ of G is defined as follows:

$$\mathfrak{g} = \{X \in M_n(\mathbb{C}) \mid e^{tX} \in G \text{ for all } t \in \mathbb{R}\}.$$

Proposition 5.1. For any matrix Lie group G , the Lie algebra $\mathfrak{g}$ of G has the following properties:

- The zero matrix 0 belongs to $\mathfrak{g}$.
- For all $X \in \mathfrak{g}$, $tX \in \mathfrak{g}$ for all real numbers t .
- For all $X, Y \in \mathfrak{g}$, $X + Y \in \mathfrak{g}$.
- For all $A \in G$ and $X \in \mathfrak{g}$ we have $AXA^{-1} \in \mathfrak{g}$.
- For all $X, Y \in \mathfrak{g}$, the commutator $[X, Y] := XY - YX$ belongs to $\mathfrak{g}$.

Proof. Write the proof here

□

Remark. Write about real vector spaces.

5.1 Later updated with the correct picture.

Chapter 6

Blocks

6.1 Equations

Here is an equation.

$$f(k) = \binom{n}{k} p^k (1-p)^{n-k} \quad (6.1)$$

You may refer to using `\@ref{eq:binom}`, like see Equation (6.1).

6.2 Theorems and proofs

Labeled theorems can be referenced in text using `\@ref{thm:tri}`, for example, check out this smart theorem 6.1.

Theorem 6.1. *For a right triangle, if c denotes the length of the hypotenuse and a and b denote the lengths of the **other** two sides, we have*

$$a^2 + b^2 = c^2$$

Read more here <https://bookdown.org/yihui/bookdown/markdown-extensions-by-bookdown.html>.

6.3 Callout blocks

The R Markdown Cookbook provides more help on how to use custom blocks to design your own callouts: <https://bookdown.org/yihui/rmarkdown-cookbook/custom-blocks.html>

Chapter 7

Sharing your book

7.1 Publishing

HTML books can be published online, see: <https://bookdown.org/yihui/bookdown/publishing.html>

7.2 404 pages

By default, users will be directed to a 404 page if they try to access a webpage that cannot be found. If you'd like to customize your 404 page instead of using the default, you may add either a `_404.Rmd` or `_404.md` file to your project root and use code and/or Markdown syntax.

7.3 Metadata for sharing

Bookdown HTML books will provide HTML metadata for social sharing on platforms like Twitter, Facebook, and LinkedIn, using information you provide in the `index.Rmd` YAML. To setup, set the `url` for your book and the path to your `cover-image` file. Your book's `title` and `description` are also used.

This `gitbook` uses the same social sharing data across all chapters in your book—all links shared will look the same.

Specify your book's source repository on GitHub using the `edit` key under the configuration options in the `_output.yml` file, which allows users to suggest an edit by linking to a chapter's source file.

Read more about the features of this output format here:

<https://pkgs.rstudio.com/bookdown/reference/gitbook.html>

Or use:

```
?bookdown:::gitbook
```